

| Question number | Answer                                                                                                                                           | Additional guidance                                                                                                                                                                          | Mark |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 9(a)(i)         | <ul style="list-style-type: none"> <li>380 and 210 / 170 / 380 and 215 / 165 (1)</li> <li>11.3 (if 210 used) / 11.0 (if 215 used) (1)</li> </ul> | <p><b>DO NOT ACCEPT 11.00 / 11.30</b></p> <p>Bald answer of {11.0 / 11.3} = 2 marks<br/> Bald answer with incorrect number of sig figs = 1 mark<br/> Bald answer of {170 / 165} = 1 mark</p> | (2)  |

| Question number | Answer                                                                                                  | Additional guidance | Mark |
|-----------------|---------------------------------------------------------------------------------------------------------|---------------------|------|
| 9(a)(ii)        | <ul style="list-style-type: none"> <li>answer in the range of 468 to 470.7 to max one dp (1)</li> </ul> |                     | (1)  |

| Question number | Answer                                                                                                                                                                     | Additional guidance                                                                                | Mark |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|------|
| 9(a)(iii)       | <p>An answer that includes the following points:</p> <ul style="list-style-type: none"> <li>(graph / line / plastic production shows a) non-linear increase (1)</li> </ul> | <p><b>ACCEPT</b> exponential rate (of increase) is not the same fluctuates increasing gradient</p> | (1)  |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                          | Additional guidance                                                                                                                                                                                                                                                                                                                       | Mark |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 9(b)(i)         | <p>A description that includes two of the following points:</p> <ul style="list-style-type: none"> <li>(the probability of death increases as the number of pieces of plastic increases in a) {S / sigmoidal} (shape) (1)</li> <li>there is a critical number of pieces of plastic above which the probability of death increases sharply (1)</li> <li>probability of death is zero when number of pieces of plastic is very low (1)</li> </ul> | <p><b>ACCEPT</b> description e.g. shallow increase then a sharp increase and then a shallow increase but non-linear positive correlation</p> <p><b>ACCEPT</b> above a certain number of pieces . . . . .</p> <p><b>ACCEPT</b> rapid increase</p> <p><b>ACCEPT</b> below a certain number of pieces of plastic the probability is zero</p> | (2)  |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Additional guidance                                                                                                          | Mark |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|------|
| 9(b)(ii)        | <p>A description that includes two of the following points:</p> <ul style="list-style-type: none"> <li>• cause of death of the turtles not known (1)</li> <li>• therefore {no direct evidence that the plastic did cause the death of the turtles / correlation not causation} (1)</li> <li>• the size of the plastic was not taken into account (1)</li> <li>• {species / size} of turtle not known (1)</li> <li>• no indication of timescale of plastic inside turtle (1)</li> <li>• no values given on the graph (1)</li> </ul> | <p><b>ACCEPT</b> other factors may have caused death</p> <p><b>ACCEPT</b> mass / type</p> <p><b>ACCEPT</b> age of turtle</p> | (2)  |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Marks |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| (c)             | <p><b>Graph 1:</b></p> <ul style="list-style-type: none"> <li>fewer mangoes are diseased (D)</li> <li>so they will be more edible / less wastage</li> <li>so more food to go round</li> <li>increase profits of mango-producing countries</li> </ul> <p><b>Graph 2:</b></p> <ul style="list-style-type: none"> <li>less mass is lost than no packaging / (slightly) more mass lost than plastic (D)</li> <li>which is probably due to dehydration of the mango</li> <li>therefore fruit will be more succulent</li> <li>and add water to the diet</li> <li>important in arid countries</li> </ul> <p><b>Table:</b></p> <ul style="list-style-type: none"> <li>decomposes relatively quickly (D)</li> <li>therefore biodegradable</li> <li>will not be any plastics to go into the sea / land fill sites</li> <li>so less damage to wild life</li> <li>less interruption to food chain</li> <li>contributes to carbon cycle</li> </ul> <p><b>Other ideas:</b></p> <ul style="list-style-type: none"> <li>sustainable / carbon neutral</li> <li>more banana plants grown reducing carbon dioxide levels</li> <li>maybe less toxic</li> <li>production of plastic results in burning of fossil fuels</li> <li>plastics not always available to countries producing mangos</li> <li>using plastic adds more to cost of mangos</li> <li>plastics are expensive</li> </ul> | <p><b>Level 1: simple</b></p> <p>1 mark = data in {one graph / table} described</p> <p>2 marks = data in {both graphs / one graph and table} described<br/> <b>OR</b><br/> one advantage described</p> <p><b>Level 2: advantages described</b></p> <p>3 marks = one advantage discussed in detail<br/> <b>OR</b><br/> two advantages described</p> <p>4 marks = two advantages discussed in detail<br/> <b>OR</b><br/> three advantages described</p> <p><b>Level 3: extended discussion</b></p> <p>5 marks = three advantages discussed in detail<br/> <b>OR</b><br/> four advantages described</p> <p>6 marks = as for 5 marks but covers 3 sections</p> | (6)   |

